

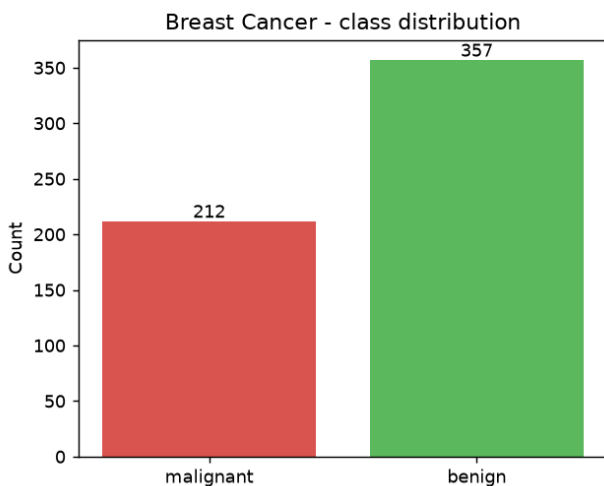
Lab 9 — Support Vector Machine (SVM) & Principal Component Analysis (PCA)

Aim: Implement SVM for classification and PCA for dimensionality reduction, and analyse their effectiveness on real-world datasets. Supervised (SVM) vs unsupervised (PCA) learning are contrasted; class balancing and stratified k-fold cross-validation are applied throughout.

Part A — Support Vector Machine (SVM)

Dataset: UCI Breast Cancer Wisconsin (Diagnostic) — 569 samples, 30 features. Features standardized (SVM is distance-based).

Class distribution



Mild imbalance: malignant 212 vs benign 357.

Method

- 80:20 train/test split with `stratify=y` (preserves class ratio).
- Class balancing via `class_weight='balanced'` to stop the minority (malignant) class being ignored.
- Hyperparameter tuning of `C` by `GridSearchCV` over a log range with 5-fold **stratified** k-fold CV, selecting the best CV F1.
- Kernel comparison: linear / rbf / poly.

Linear SVM — baseline (C=1) vs balanced

Metric	Balanced (C=1)
Accuracy	0.9561
Precision	0.9855
Recall	0.9444
F1 Score	0.9645

Confusion matrix (rows=true, cols=pred): $\begin{bmatrix} 41 & 1 \\ 4 & 68 \end{bmatrix}$

Tuned linear SVM (best C)

GridSearchCV → **best C = 0.0100**, mean CV F1 = **0.9808** (5-fold stratified CV).

Metric	Test value	
Accuracy	0.9737	0.9737
Precision	0.9859	0.9859
Recall	0.9722	0.9722
F1 Score	0.9790	0.9790

Confusion matrix: $\begin{bmatrix} 41 & 1 \\ 2 & 70 \end{bmatrix}$

Stratified k-fold cross-validation (full data, 5 folds)

Score	Mean ± Std
Accuracy	0.9772 ± 0.0089
F1	0.9820 ± 0.0069
Precision	0.9757 ± 0.0195
Recall	0.9888 ± 0.0162

Kernel comparison (CV F1 over 5-fold stratified CV)

Kernel	CV F1	Test Accuracy	Test F1
linear	0.9755	0.9561	0.9645
rbf	0.9807	0.9649	0.9718
poly	0.9409	0.9561	0.9664

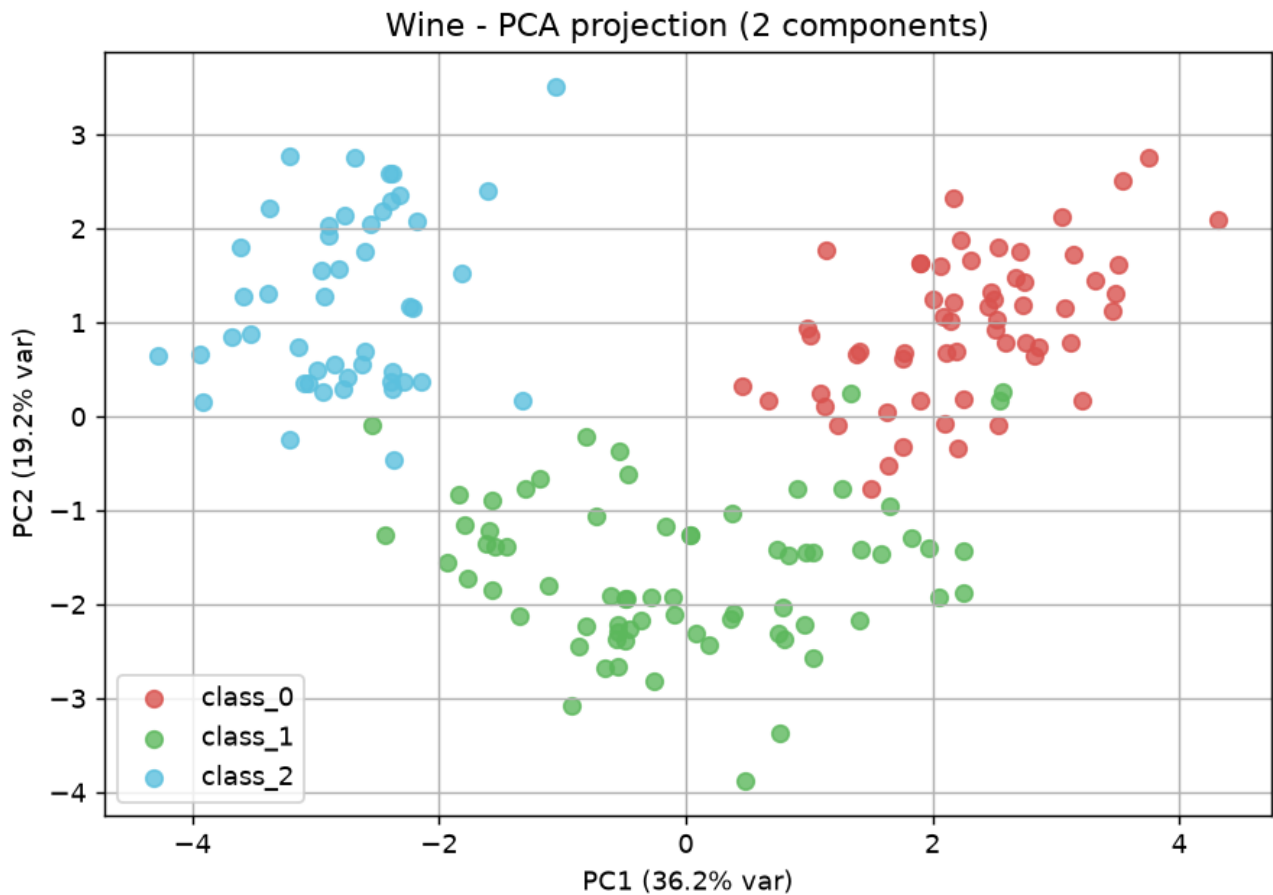
Winner (CV F1): rbf.

Part A observations. The classes are well separated after standardization, so every kernel scores high. Class balancing lifts recall on the minority class; stratified CV shows stable generalization (low std). The tuned C (0.0100) is small, indicating a wide margin is preferred. rbf edges out linear and poly on CV F1 for this scaled data.

Part B — Principal Component Analysis (PCA)

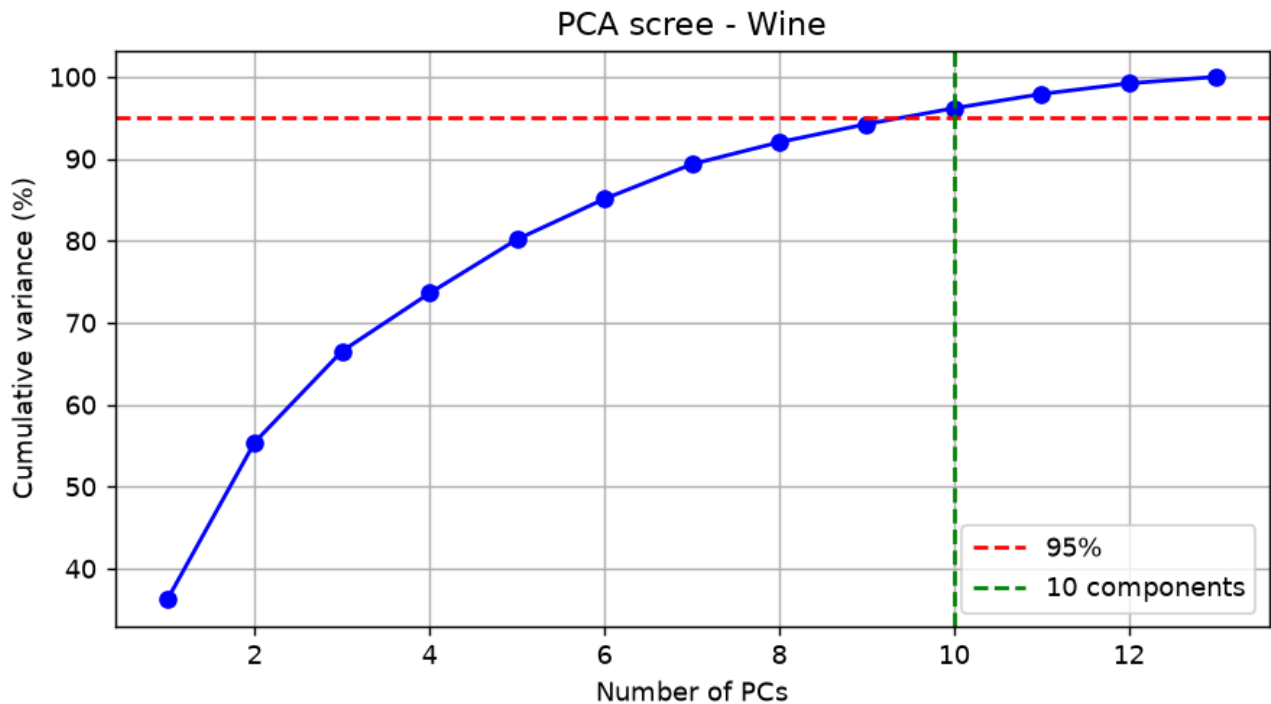
Dataset: UCI Wine — 178 samples, 13 features (3 cultivars). Features standardized so no single variable dominates the variance.

PCA projection & explained variance



2-D PCA projection of Wine; PC1 = 36.2% variance, PC2 = 19.2% (total 55.4% retained in 2 PCs).

Variance retained & minimum components for $\geq 95\%$



The cumulative curve crosses 95% variance at **10 of 13** components — a strong compression: from 13 dimensions down to 10 keeps essentially all information.

Interpretation of PC1 & PC2 (top loadings)

Component	Top contributing features (weight)
PC1	flavanoids (+0.42), total_phenols (+0.39), od280/od315_of_diluted_wines (+0.38), proanthocyanins (+0.31), nonflavanoid_phenols (-0.30)
PC2	color_intensity (+0.53), alcohol (+0.48), proline (+0.36), ash (+0.32), magnesium (+0.30)

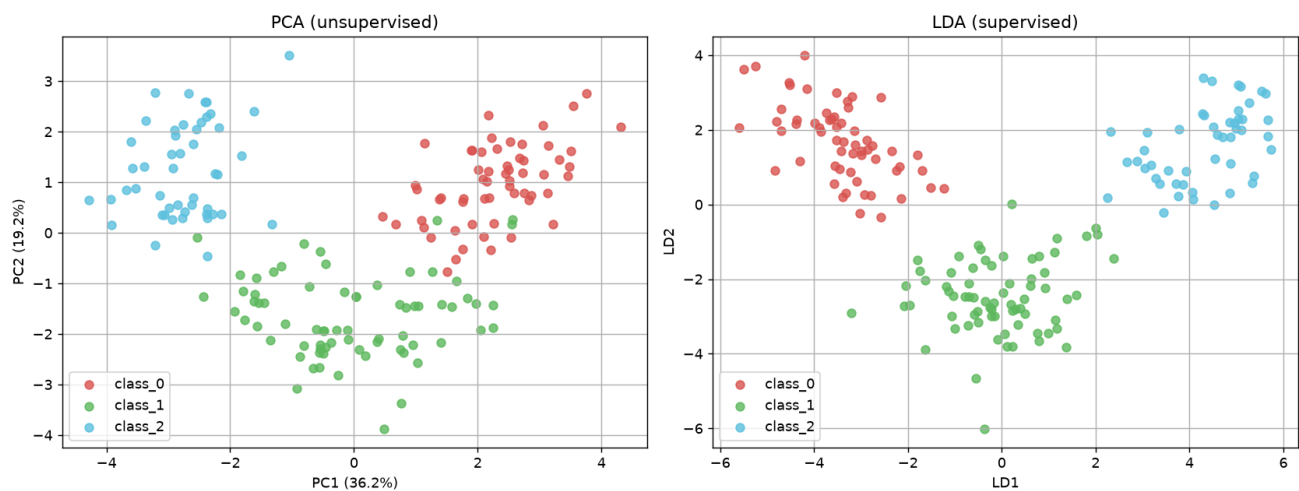
Original vs transformed

Aspect	Original (13-D)	PCA (2-D)
# features	13	2
Information	100%	55.4% retained
Efficiency	slower	faster
Separability	full	mostly kept
Interpretability	per-feature	mixed combos

PCA pros: decorrelates, denoises, compresses, speeds models, enables 2-D visualization. **Cons:** loses variance, components are harder to interpret, scale-sensitive, unsupervised (ignores labels).

Applications: preprocessing, denoising, eigenfaces, visualization, factor analysis.

Extra credit — Linear Discriminant Analysis (LDA)



LDA explained-variance ratio: $[0.6875, 0.3125]$ (total 100.00%). LDA is **supervised** — it uses class labels to maximize separation, so the three cultivars form tighter clusters than in PCA. Trade-off: LDA needs labels and is limited to $C-1 = 2$ components for 3 classes.

Conclusions

1. **SVM (supervised):** linear SVM with `class_weight='balanced'` handled the mild imbalance; stratified k-fold CV confirmed stable generalization; kernel chosen by CV F1 (winner: rbf).
2. **PCA (unsupervised):** reduced Wine 13→2 retaining 55.4% variance; only **10** components needed for $\geq 95\%$ variance — strong compression.
3. **Supervised vs unsupervised:** SVM uses labels to classify; PCA ignores labels and only compresses variance. LDA (extra credit) uses labels and yields tighter class clusters than PCA in 2-D, at the cost of needing labels and fewer components.

All results use `random_state=42`, `class_weight='balanced'`, and stratified 5-fold CV for reproducibility.